Project: Monte Carlo simulation of one-dimensional systems

Objective: To study the structure of one-dimenssional systems with different interaction potentials.

Outline

1. Write a MC code for a simulation of a Lennard-Jones (LJ) system in one dimension. Use N = 100 atoms. Remember that the LJ potential is of the form: $U_{LJ}(r) = 4\varepsilon[(\sigma/r)^{12} - (\sigma/r)^6]$.
2. Create an initial set-up of positions from the sites of a one-dimensional lattice.
3. Run a MC simulation for argon at ρ^*= 0.95 and T*=1.0. Compute the interaction energy U^*, the pressure P^* and the radial distribution function $g(r)$ of the system. Compute the running coordination number $(n(r))$ via integration of the g(r).
4. Repeat the simulation for the temperatures T*= 2.0 and and T*= 4.0 , and compare with the previous results.
5. Repeat thes ame study for $\rho^* = 0.65$ and temperatures T* = 1.0, T* = 2.0, and T* = 4.0.
6. Repeat the simulations for a system of hard spheres, with an interaction potential given by

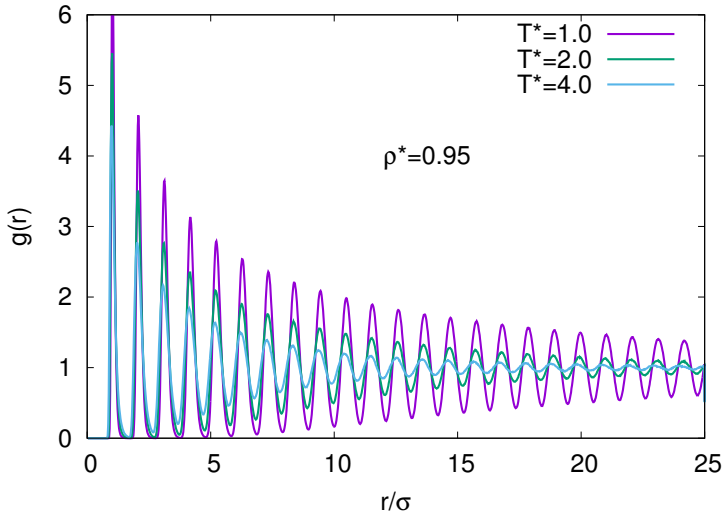
$$U_{hs}(r) = \begin{cases} \infty, & 0 < r \leq \sigma \\ 0, & r > \sigma \end{cases}$$

   Compute the radial distribution function and compare it with those obtained with the Lennard-Jones potential.

Lennard-Jones parameters

atom	mass(uma)	ϵ/k_B (K)	σ ($\mathring{A}$)
Ar	39.948	119.8	3.405

Some results to compare with:



Radial distribution functions for the Lennard-Jones system at $\rho^* = 0.95$ and different temperatures.

Questions

1. Give the expressions of the pressure P and the radial distribution $g(r)$ for the case of a one-dimensional system.

2. Derive the expressions for the tail corrections to the energy (U_{LJ}^{tail}) and pressure (P_{LJ}^{tail}).

   HINT: Remember that for a three dimensional system it is

   $$U = \frac{1}{2} N \rho \int_0^\infty U(r) g(r) 4\pi r^2 dr$$

   and

   $$P = \rho k_B T - \frac{\rho^2}{6} \int_0^\infty r \frac{dU(r)}{dr} g(r) 4\pi r^2 dr$$

3. Define the reduced units used in your simulations. Translate from reduced units to real units the values of T^*, ρ^*, U^* and P^*.

4. Describe the implementation of the Metropolis algorithm for the hard spheres system.